

SURVEY-WEIGHTED ORDINAL CAUSAL DISCOVERY: APPLICATION TO VETERAN SOCIAL NEEDS, MENTAL HEALTH, AND WELL-BEING

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Social determinants of health (SDOH) are highly intercorrelated, making it difficult to differentiate root causes from mediating factors using standard associational methods. While causal discovery algorithms offer a data-driven approach to learn such causal structures, existing methods fail when applied to survey data. They rely on the assumption of independent and identically distributed data, whereas modern surveys employ unequal selection probabilities. To overcome this limitation for ordinal survey data, we introduce the survey-weighted ordinal Bayesian network, or *swa-OBn*. By integrating normalized Horvitz–Thompson weights and Kish’s effective sample size into a pseudo-BIC score for ordinal Bayesian networks and exploiting an asymmetry of the ordinal cumulative link parameterization, *swa-OBn* generically identifies causal directionality within Markov equivalence classes. We prove that the framework consistently recovers the true population-level causal graph in large samples, even under complex survey designs. We apply *swa-OBn* to national Veterans Affairs (VA) CAHPS survey data to map the causal architecture of Veteran SDOH and health outcomes. We demonstrate that ignoring the complex survey design misrepresents discrimination as a root cause of material deprivation and health outcomes. By explicitly modeling the sampling design, *swa-OBn* recovers the true structural hierarchy, revealing that unmet basic needs are the actual upstream hub, with discrimination acting solely as a downstream mediator to poor mental health. Complementary simulations confirm that ignoring complex survey designs produces severely biased causal graphs.

1. Introduction. Social determinants of health (SDOH) such as financial instability, housing insecurity, and social isolation are highly intercorrelated in population surveys, creating a fundamental challenge for intervention design. Standard regression analyses document associations between unmet social needs and poor health outcomes (Braveman, Egerter and Williams, 2011; Duan-Porter et al., 2018), but they cannot differentiate upstream root causes from downstream mediating pathways (Glass et al., 2013; Athey, 2017). This distinction is critical for policy: if unmet basic needs are a root cause that cascades into discrimination, isolation, and poor mental health, then interventions targeting basic needs will have broad downstream benefits; if the causal ordering is different, so is the optimal allocation of resources.

Causal discovery algorithms offer a data-driven methodology for inferring directed causal structure from observational data (Spirtes, Glymour and Scheines, 2000; Pearl, 2009; Glymour, Zhang and Spirtes, 2019). Unlike structural equation modeling (Wright, 1921), which

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requires the causal architecture to be pre-specified, these algorithms learn the directed acyclic graph (DAG) from the data. However, applying causal discovery to survey data introduces a methodological problem that, to our knowledge, has not been addressed in the literature: complex surveys produce non-i.i.d. data, and ignoring the survey design yields distorted or biased underlying causal relationships.

To illustrate this problem concretely, consider a simple population causal chain $X \rightarrow Y \rightarrow Z$, where X , Y , and Z are ordinal variables representing basic financial need, access to resources, and well-being. In the population, X and Z are conditionally independent given Y . Now suppose a survey over-samples individuals with both high X (severe need) and low Z (poor well-being). The resulting sample over-represents the joint tail of (X, Z) , inducing a spurious marginal dependence between X and Z that persists even after conditioning on Y . An unweighted causal discovery algorithm, confronted with this distorted distribution, will infer a direct edge $X \rightarrow Z$ (or $Z \rightarrow X$) that does not exist in the population, producing a causal graph that reflects the sampling mechanism rather than the data-generating process. This bias is systematic, not random: it worsens with stronger oversampling and does not vanish as the sample size increases.

The problem is compounded by a second challenge specific to categorical data. Standard multinomial Bayesian networks (BNs) are non-identifiable: multiple DAGs that encode the same conditional independence relationships produce identical likelihoods (i.e., Markov equivalence), so the direction of many edges cannot be determined even with infinite data (Heckerman, Meek and Cooper, 1997; Chickering, 2002). Ni and Mallick (2022) resolved this identifiability problem for ordinal data by introducing the ordinal Bayesian network ($\circ$ BN). They proved that the ordinal parameterization generically breaks Markov equivalence when all variables have three or more levels, enabling full DAG identifiability from observational data. Ni, Chen and Wang (2025) extended this to a bootstrapped multivariate algorithm with a BIC-based score and demonstrated its utility in a study of obsessive-compulsive disorder and depression. To our knowledge no other causal discovery algorithm achieves full DAG identifiability for a network of ordinal data. However, the $\circ$ Bn framework assumes i.i.d. data, does not support exogenous covariate adjustment, and uses a single BIC score that introduces a finite-sample orientation bias when ordinal variables have heterogeneous numbers of levels.

This paper addresses these gaps by introducing the *survey-weighted, covariate-adjusted ordinal Bayesian network*, or $\text{swa-}\circ\text{Bn}$. The method makes four contributions over the existing $\circ$ Bn framework:

1. *Survey weighting.* We integrate normalized Horvitz–Thompson weights (Horvitz and Thompson, 1952) and Kish’s effective sample size (Kish, 1965) into a pseudo-BIC scoring framework, enabling design-consistent structure learning from complex survey data.
2. *Covariate adjustment.* Measured demographic confounders are absorbed into every node-wise regression without expanding the DAG, reducing confounding bias while preserving the ordinal identifiability guarantee.
3. *Scoring variants for heterogeneous ordinal scales.* We characterize the existing single-BIC score of Ni and Mallick (2022); Ni, Chen and Wang (2025) as one of two co-equal variants and introduce a second, *split-BIC*, that scores edge reversals on the unpenalized pseudo-likelihood while retaining the pseudo-BIC for additions and deletions. Both variants are large-sample consistent. Under heterogeneous numbers of ordinal levels, neither dominates: their relative finite-sample accuracy is governed by the structure of the level vector $\mathbf{L} = (L_1, \dots, L_q)$, and we provide a regime-based decision rule supported by empirical evidence (see Section 6).
4. *Structural constraints.* Domain knowledge is encoded via blacklisted and whitelisted edges, restricting the search space to scientifically plausible graphs.

We prove that the `swa-OBn` consistently recovers the true population-level DAG in large samples under a joint superpopulation-and-design asymptotic framework, with skeleton consistency established via the penalized pseudo-BIC and orientation consistency inherited by both scoring variants from generic ordinal identifiability.

We apply the `swa-OBn` to national data from the Veterans Health Administration (VHA) Consumer Assessment of Healthcare Providers and Systems (CAHPS) survey to map the causal architecture of social determinants of health among U.S. Veterans. Recent analyses of this dataset have documented substantial variation in social need prevalence (Frank et al., 2025) across age and race-ethnicity-sex, and shown that standard social risk screening under-detects needs among certain demographic subgroups (Russell et al., 2026). These analyses rely on associational methods that assume rather than discover causal structure. The `swa-OBn` addresses this gap, revealing that unmet basic needs act as an upstream causal hub, while discrimination and isolation serve as the proximal mediating pathway to poor mental health and well-being.

The remainder of this paper is organized as follows. Section 2 introduces the CAHPS data and the scientific questions motivating the analysis. Section 3 reviews ordinal Bayesian networks. Section 4 develops the `swa-OBn` methodology. Section 5 establishes theoretical properties. Section 6 presents simulation studies. Section 7 reports the application to VHA data. Section 8 concludes with a discussion. An R implementation of the proposed methodology, the `swaOBn` package, is available with full documentation online.

2. Data and Scientific Questions.

2.1. The Veterans Health Administration and social determinants of health. The Veterans Health Administration (VHA) is the largest integrated health care system in the United States, providing care to over nine million enrolled Veterans annually (U.S. Department of Veterans Affairs, 2026). Understanding and addressing social determinants of health (SDOH) among this population has become a major policy objective (Daniel, Bornstein and Kane, 2021). The existing literature on SDOH in the Veteran population has extensively documented that unmet social needs are associated with poor health outcomes (Braveman, Egerter and Williams, 2011; Duan-Porter et al., 2018). Recent analyses of the VHA patient population have revealed substantial variation in social need prevalence: compared with White male Veterans, Black male Veterans report elevated needs in 11 of 13 domains, and need for support managing discrimination is elevated across all non-White-male subgroups (Frank et al., 2025; Lamba et al., 2025). Furthermore, standard social risk screening instruments under-detect need for support among certain demographic subgroups, particularly Black male Veterans (Russell et al., 2026). These findings confirm that SDOH are pervasive and heterogeneous, but the analyses underlying them are entirely associational. Regression models report which needs are correlated with which outcomes; they do not reveal the causal ordering among needs, leaving policymakers without the directional knowledge required to prioritize interventions effectively. This paper addresses the following policy questions that standard associational methods cannot answer:

1. Which unmet social needs are upstream root causes of other needs, and which are downstream consequences?
2. Among the needs causally linked to poor health, which function as *distal* causes (i.e., operating on outcomes indirectly through mediating needs) and which function as *proximal* causes that transmit upstream effects directly to health?

The distal–proximal distinction is not merely descriptive; it is a property of the causal topology that only a DAG-based analysis can reveal. A need that is strongly correlated with

mental health may be causally distal (acting through intermediate mediators, so that interventions bypass most of its effect unless the mediators are also addressed) or causally proximal (directly adjacent to the outcome, so that targeting it yields immediate benefit). Associational methods cannot distinguish these cases; the `swa-oBn` can.

2.2. Survey design and data. We utilize data from the Consumer Assessment of Healthcare Providers and Systems (CAHPS) Clinician and Group Visit Survey, a standardized national instrument developed by the Agency for Healthcare Research and Quality (AHRQ) and administered by the VHA as part of its Survey of Healthcare Experiences of Patients (SHEP) program (Dyer et al., 2012; Agency for Healthcare Research and Quality, 2024; Hausmann et al., 2023). The survey was fielded to a stratified random sample of VHA primary care patients seen in January or February 2023. To ensure adequate statistical power for inter-sectional comparisons, the design deliberately oversampled women and Black and Hispanic Veterans, producing six race-ethnicity-sex strata with approximately 900 respondents each (Frank et al., 2025). The analytic sample comprises $n = 5,396$ respondents (complete-case analysis). Survey weights account for the sampling design and survey nonresponse and vary substantially: the ratio of the largest to smallest weight is approximately 990:1, with a coefficient of variation of 1.70. Kish’s effective sample size is $n_{\text{eff}} = 1,387$, yielding a design effect of 3.89. This effective sample size governs the penalty term in the pseudo-BIC (Section 4.2) and reflects a nearly fourfold reduction in effective information relative to a simple random sample of the same size.

2.3. Variable construction. The CAHPS survey assesses need for support across 13 social domains using a three-level ordinal scale: “no support needed” (level 1), “needed support and got it” (level 2), and “needed support but did not get it” (level 3). Higher values represent more severe unmet need. The companion analyses of Frank et al. (2025) and Russell et al. (2026) collapsed levels 2 and 3 into a binary indicator of need for the purpose of estimating prevalence ratios. The present analysis retains the full ordinal scale, which is essential: the identifiability result of Ni and Mallick (2022) that underlies our framework requires variables with three or more levels, and the three-level response captures the substantively important distinction between met and unmet need.

A fundamental challenge in applying causal discovery to survey instruments is the high dimensionality and severe collinearity of item-level data. To construct conceptually distinct nodes for the Bayesian network, we analyzed the item-level polychoric correlation structure. Items exhibiting polychoric correlations $\rho \geq 0.75$ were aggregated into composite ordinal domains by taking the most severe level of need across the constituent items, thus preserving the original three-level ordinal scale. This yielded three composite nodes: *basic needs* (food, housing, paying for basics; $\rho = 0.85\text{--}0.95$), *work/education* (finding work, additional education; $\rho = 0.82$), and *discrimination/isolation* (discrimination, social isolation, loneliness; $\rho = 0.75\text{--}0.96$). Five items falling below this threshold were retained as standalone nodes: *caregiving*, *childcare*, *transportation*, *internet access*, and *legal assistance*. The downstream health outcomes were captured by two additional ordinal nodes: *well-being* (four levels) and *mental health* (five levels). The complete node set consists of $q = 10$ ordinal variables with 3–5 levels each. The polychoric correlation matrix and clustering are presented in the Supplementary Material.

REMARK 1. *The polychoric correlation threshold $\rho \geq 0.75$ was chosen in collaboration with domain experts to balance construct validity with network interpretability. We treat this as a deliberate methodological trade-off, accepting an approximate working model in exchange for an interpretable graph. The bootstrap procedure of Section 4.4 provides robustness against orientation artifacts that might arise from this approximation. A fuller discussion of the implications of this choice appears in Section 8.*

2.4. *Covariate adjustment and structural constraints.* Observational causal discovery is subject to the assumption of causal sufficiency: all common causes of any pair of variables in the DAG must be measured. To mitigate confounding by demographic characteristics, our analysis adjusts for age (categorized into four bands) and a combined race-ethnicity-sex indicator, the same covariates used in the companion analyses of this dataset (Frank et al., 2025; Lamba et al., 2025; Russell et al., 2026). These variables are not treated as nodes in the DAG; instead, they form an exogenous covariate matrix $\mathbf{Z}$ that is conditioned on in every node-wise regression (Section 4.3), ensuring the discovered edges represent relationships adjusted for demographic confounding. A discussion of what $\mathbf{Z}$ does and does not absorb is deferred to Section 8.

Finally, we integrate domain knowledge via a structural blacklist. All directed edges originating from the two outcome nodes (well-being and mental health) and pointing toward any of the eight social need nodes are forbidden. This encodes the minimal substantive assumption that structural social needs are upstream of health states. All relationships among the social need variables themselves are left entirely unconstrained.

3. Background: Ordinal Bayesian Networks. A Bayesian network (Bn) represents the joint distribution of random variables $\mathbf{X} = (X_1, \dots, X_q)$ through a directed acyclic graph (DAG) $\mathcal{G} = (V, E)$ and an associated factorization $P(\mathbf{X} | \mathcal{G}) = \prod_{j=1}^q P(X_j | \mathbf{X}_{\text{pa}(j)})$, where $\text{pa}(j)$ denotes the parent set of node j in $\mathcal{G}$ (Pearl, 2009). Under the standard assumptions of causal sufficiency and faithfulness (see Section 5.1) the edges of $\mathcal{G}$ admit a causal interpretation and the DAG can be recovered from observational data. For categorical variables, each conditional $P(X_j | \mathbf{X}_{\text{pa}(j)})$ may be specified by a multinomial distribution. A well-known limitation of this parameterization is that it is identifiable only up to a Markov equivalence class: all DAGs encoding the same conditional independence relations produce the same likelihood, leaving the direction of many edges undetermined (Heckerman, Meek and Cooper, 1997; Chickering, 2002).

When the categorical variables are ordinal, the multinomial parameterization discards informative structure. Ni and Mallick (2022) proposed the ordinal Bayesian network (oBn), which replaces multinomial conditionals with cumulative link regressions. For an ordinal variable X_j with L_j levels and parent set $\text{pa}(j)$, the oBn specifies

$$(1) \quad \Pr(X_j \leq \ell | \mathbf{X}_{\text{pa}(j)}) = \Phi \left(\gamma_{j\ell} - \sum_{k \in \text{pa}(j)} \beta_{jk} X_k - \alpha_j \right), \quad \ell = 1, \dots, L_j,$$

where Φ is the standard normal cumulative distribution function (any strictly increasing link F may be used; we adopt Φ for concreteness and consistency with Ni and Mallick (2022); Ni, Chen and Wang (2025)), $\gamma_{j1}, \dots, \gamma_{j, L_j-1}$ are ordered cutpoints with $\gamma_{j1} = 0$ for identifiability, α_j is an intercept, and β_{jk} captures the effect of parent X_k on X_j . When a parent X_k enters as a predictor with L_k levels, separate coefficients $\beta_{jk1}, \dots, \beta_{jk, L_k-1}$ replace the scalar β_{jk} , with $\beta_{jk1} = 0$ as a reference constraint. The total count of free parameters under DAG $\mathcal{G}$ is

$$(2) \quad K_{\mathcal{G}} = \sum_{j=1}^q \left(L_j - 1 + \sum_{k \in \text{pa}(j)} (L_k - 1) \right).$$

This count plays a role in the BIC penalty and will be important for motivating the single- and split-BIC scoring procedure of Section 4.1.

The central property of the oBn is generic identifiability. Ni and Mallick (2022) proved that if $X_j \rightarrow X_k$ is the true data-generating direction and both variables have three or more

levels, then, except on a Lebesgue-null subset of the parameter space, the reverse model $X_k \rightarrow X_j$ cannot reproduce the same joint distribution under the ordinal parameterization. That is, the sets of joint distributions reachable under $X_j \rightarrow X_k$ and $X_k \rightarrow X_j$ are generically distinct. Ni, Chen and Wang (2025) extended this to the multivariate setting and introduced a BIC-based greedy hill-climbing algorithm with bootstrap uncertainty quantification. This identifiability result is what makes score-based structure learning viable for ordinal data: the true DAG can be distinguished from all Markov-equivalent alternatives, not merely located within an equivalence class. However, the framework of Ni and Mallick (2022); Ni, Chen and Wang (2025) assumes i.i.d. data and does not accommodate survey weights or exogenous covariates. The next section develops the extensions required to address these limitations.

4. Survey-Weighted Ordinal Causal Discovery. We now develop the survey-weighted, covariate-adjusted ordinal Bayesian network ($\text{swa-}\text{oBn}$), extending the oBn of Ni and Mallick (2022); Ni, Chen and Wang (2025) to complex survey data with measured confounders and domain knowledge. The method has four components: a node-wise scoring framework with two variants, single-BIC and split-BIC (Section 4.1); survey weighting (Section 4.2); covariate adjustment (Section 4.3); and bootstrap uncertainty quantification (Section 4.4). Section 4.5 closes with the choice of variant for the application. Algorithm pseudocode is in the Supplementary Material.

4.1. Node-wise scoring: single-BIC and split-BIC. Let $\mathbf{x} = (x_1, \dots, x_n)$ denote n realizations of $\mathbf{X}$. Under the oBn factorization with cumulative probit conditionals (1), the log-likelihood decomposes node-wise,

$$(3) \quad \ell(\boldsymbol{\theta}; \mathcal{G}) = \sum_{j=1}^q \ell_j(\boldsymbol{\theta}_j; \text{pa}(j)), \quad \ell_j(\boldsymbol{\theta}_j; \text{pa}(j)) = \sum_{i=1}^n \log p(x_{ij} | \mathbf{x}_{i, \text{pa}(j)}, \boldsymbol{\theta}_j),$$

with $\boldsymbol{\theta}_j = (\gamma_j, \beta_j, \alpha_j)$. Ni and Mallick (2022); Ni, Chen and Wang (2025) score candidate graphs by

$$(4) \quad \text{BIC}(\mathcal{G}) = -2\ell(\hat{\boldsymbol{\theta}}; \mathcal{G}) + K_{\mathcal{G}} \log n,$$

where $K_{\mathcal{G}}$ is given by (2), and select the DAG minimizing this score.

Exhaustive enumeration is infeasible: the number of DAGs on q nodes is super-exponential in q , and every graph requires fitting a node-wise cumulative link regression for each parent set. Score-based structure learning therefore proceeds by greedy hill-climbing (Chickering, 2002; Scutari, 2010), which navigates the DAG space through local moves. From an initial graph, each iteration considers three move types subject to acyclicity: additions and deletions, which alter the skeleton, and reversals, which flip the direction of an existing edge. The node-wise decomposition (3) makes this efficient: a single-edge modification changes the parent set of at most two nodes, so only those node-wise scores need recomputation. The search terminates at a local optimum; we use R random restarts and return the solution with the lowest penalized score.

Reversals are the consequential case for scoring: they navigate within Markov equivalence classes, where ordinal identifiability (Ni and Mallick, 2022) is the only mechanism distinguishing $k \rightarrow j$ from $j \rightarrow k$. When all variables share a common number of levels L , the parameter counts of $k \rightarrow j$ and $j \rightarrow k$ coincide and the BIC penalties cancel within any equivalence class; orientation is governed entirely by the likelihood. When ordinal scales are heterogeneous, the regression of X_j on X_k contributes $L_k - 1$ slopes while the reverse contributes $L_j - 1$, so Markov-equivalent DAGs differing only in one edge direction receive different BIC penalties. This admits two resolutions, which we present as alternative scoring variants:

- *Single-BIC scoring* (Ni and Mallick, 2022; Ni, Chen and Wang, 2025) applies (4) uniformly to all three move types. Under heterogeneous L , the penalty differential between $k \rightarrow j$ and $j \rightarrow k$ is $(L_k - L_j) \log n$, biasing reversal decisions toward making the lower- L variable the parent. This bias accumulates across edges and may hurt overall orientation accuracy.
- We develop *Split-BIC scoring* that applies (4) to additions and deletions but evaluates reversals on the unpenalized $-2\ell(\boldsymbol{\theta}; \mathcal{G})$, removing the penalty differential and reducing the within-class comparison to a likelihood ratio. By generic ordinal identifiability, the correctly oriented edge attains a strictly higher likelihood for almost all parameter values, regardless of L .

The two variants coincide under homogeneous L . Under heterogeneous L , neither dominates: relative performance depends on the structure of $L = (L_1, \dots, L_q)$, with empirical evidence in Section 6.5. Both variants share the same large-sample guarantees; see Section 5.

REMARK 2 (Structural constraints). *A blacklist $\mathcal{B}$ of forbidden directed edges and a whitelist $\mathcal{W}$ of required directed edges encode domain knowledge. Blacklisted edges are never proposed and reversals creating one are skipped; whitelisted edges are placed in the initial graph and locked.*

4.2. *Survey weighting.* When inclusion probabilities are associated with the variables under study, the sampling mechanism is informative (Pfeffermann, 1993) and parameter estimates obtained by treating the sample as i.i.d. can be inconsistent for the corresponding population quantities. The graph learned from an unweighted analysis may then reflect the sampling design rather than the population-generating causal structure.

The standard correction is pseudo-maximum likelihood estimation (Binder, 1983; Skinner, 1989). Let S denote a probability sample of size n drawn from a finite population of size N , with first-order inclusion probabilities $\pi_i = \Pr(i \in S)$ and survey weights $w_i = 1/\pi_i$. We replace the log-likelihood (3) with the pseudo-log-likelihood

$$(5) \quad \ell^w(\boldsymbol{\theta}; \mathcal{G}) = \sum_{i \in S} w_i \log p(\mathbf{x}_i | \mathcal{G}, \boldsymbol{\theta}),$$

which is design-unbiased for the census log-likelihood by the Horvitz–Thompson property. The node-wise factorization of (3) is preserved. The pseudo-BIC replacing (4) is

$$(6) \quad \text{pBIC}(\mathcal{G}) = -2\ell^w(\hat{\boldsymbol{\theta}}^w; \mathcal{G}) \frac{n_{\text{eff}}}{n} + K_{\mathcal{G}} \log(n_{\text{eff}}),$$

where $\hat{\boldsymbol{\theta}}^w$ is the pseudo-maximum likelihood estimate and $n_{\text{eff}} = (\sum_{i \in S} w_i)^2 / \sum_{i \in S} w_i^2$ is Kish’s effective sample size (Kish, 1965). With weights normalized to $\sum_i w_i = n$, the factor n_{eff}/n rescales the raw pseudo-log-likelihood from the sample scale to the effective-information scale, matching the penalty (Lumley and Scott, 2015). When all weights are equal, $n_{\text{eff}} = n$ and (6) reduces to (4).

The two scoring variants of Section 4.1 take their survey-weighted forms by replacing (4) with (6) for additions and deletions. For reversal moves, single-BIC retains the pseudo-BIC; split-BIC drops the penalty and uses the rescaled pseudo-log-likelihood $-2\ell^w(\hat{\boldsymbol{\theta}}^w; \mathcal{G})(n_{\text{eff}}/n)$.

REMARK 3 (Relation to the theoretical pseudo-BIC). *The pseudo-BIC (6) uses normalized weights ($\sum_{i \in S} w_i = n$) with likelihood rescaling n_{eff}/n and penalty $\log(n_{\text{eff}})$; Section 5 analyzes the equivalent formulation with raw weights ($w_i = 1/\pi_i$) and penalty ρ_n . The ratio N/n_{eff} is graph-invariant, so $\arg \min_{\mathcal{G}} \text{pBIC}(\mathcal{G})$ coincides under either parameterization.*

4.3. *Covariate adjustment.* Measured covariates $\mathbf{Z} = (Z_1, \dots, Z_r) \in \mathbb{R}^r$ may confound relationships among $\mathbf{X}$ but are not themselves objects of causal discovery. We absorb their influence by extending (1) to include $\mathbf{Z}$ as an additive predictor at every node:

$$(7) \quad \Pr(X_j \leq \ell \mid \mathbf{X}_{\text{pa}(j)}, \mathbf{Z}) = \Phi \left(\gamma_{j\ell} - \sum_{k \in \text{pa}(j)} \beta_{jk} X_k - \mathbf{Z}' \boldsymbol{\delta}_j - \alpha_j \right), \quad \ell = 1, \dots, L_j,$$

where $\boldsymbol{\delta}_j$ is node-specific. Crucially, $\mathbf{Z}$ does not appear as a node in $\mathcal{G}$: the search adds, deletes, and reverses edges only among $\mathbf{X}$, so the search space remains $O(q^2)$ rather than $O((q+r)^2)$. The pseudo-BIC and scoring framework of Sections 4.1–4.2 apply unchanged with $\boldsymbol{\delta}_j$ absorbed into the parameter vector. Under this extension the causal sufficiency assumption applies to the enlarged variable set: all common causes of any pair (X_j, X_k) must lie in $\mathbf{X} \cup \mathbf{Z}$ rather than $\mathbf{X}$ alone. Adding measured confounders to $\mathbf{Z}$ therefore weakens the substantive demands placed on the analyst. Generic identifiability is preserved: for each fixed $\mathbf{Z}$, model (7) reduces to an oBN with shifted intercepts, so Ni and Mallick (2022) continues to hold (Proposition 1, Section 5).

4.4. *Bootstrap uncertainty quantification.* A single hill-climbing run produces a point-estimate DAG $\hat{\mathcal{G}}$. To quantify structural uncertainty we use the nonparametric bootstrap (Efron and Tibshirani, 1993). For each replicate $b = 1, \dots, B$, we resample n tuples $(\mathbf{x}_i, w_i, \mathbf{z}_i)$ with replacement, recompute n_{eff} from the resampled weights, and re-run the search under the chosen scoring variant to obtain $\hat{\mathcal{G}}^{(b)}$. The edge inclusion probability for $j \rightarrow k$ is

$$(8) \quad \hat{p}_{jk} = B^{-1} \sum_{b=1}^B \mathbf{1}(j \rightarrow k \in \hat{\mathcal{G}}^{(b)}),$$

and a consensus graph retains edges with $\hat{p}_{jk} \geq 0.50$. This adapts the procedure of Ni, Chen and Wang (2025) to the survey-weighted setting; pseudocode is in the Supplementary Material.

4.5. *Choosing between single-BIC and split-BIC.* The two scoring variants of Section 4.1 coincide under homogeneous ordinal scales. Under heterogeneous scales, finite-sample performance depends on the structure of the level vector $\mathbf{L} = (L_1, \dots, L_q)$, with neither variant dominating across all configurations; Section 6 presents a systematic comparison. For the application in Section 7, the SHEP level vector $\mathbf{L} = (3, 3, 3, 3, 3, 3, 3, 3, 4, 5)$ falls in the regime where single-BIC outperforms split-BIC, and we use single-BIC throughout.

5. Theoretical Properties. We establish the large-sample properties of the swa-oBN under a joint asymptotic framework that accounts for two sources of randomness: the superpopulation ξ generating the finite population, and the sampling design selecting units from it. Let $U_N = \{1, \dots, N\}$ denote a finite population of size N , generated as N independent draws from ξ , with true causal DAG $\mathcal{G}^*$ and parameters $\boldsymbol{\theta}^*$. From U_N , a probability sample S_n of size $n = n(N)$ is drawn according to a sampling design $p_N(S)$ with known first-order inclusion probabilities $\pi_i = \Pr(i \in S_n)$ and survey weights $w_i = 1/\pi_i$. We require $N \rightarrow \infty$ and $n \rightarrow \infty$ with sampling fraction $n/N \rightarrow f \in [0, 1)$. All convergence statements are under the joint $(\xi \times p_N)$ -probability. Proofs are given in the Supplementary Material.

We use $\boldsymbol{\theta}_{\mathcal{G}}^*$ to denote the superpopulation maximizer of $Q_{\xi}(\cdot \mid \mathcal{G})$, i.e., $\boldsymbol{\theta}_{\mathcal{G}}^* = \arg \max_{\boldsymbol{\theta}} Q_{\xi}(\boldsymbol{\theta} \mid \mathcal{G})$; when $\mathcal{G} = \mathcal{G}^*$, this coincides with the true data-generating parameter, which we denote

by θ^* . For a mis-specified DAG $\mathcal{G}' \neq \mathcal{G}^*$, we use $\theta_{\mathcal{G}'}^\dagger \equiv \theta_{\mathcal{G}'}^*$ to emphasize its role as the KL-projection rather than a truth-level parameter. The *conditional* (covariate-pointwise) KL-projection is denoted $\theta^\dagger(z) = \arg \max_{\theta} \sum_{\mathbf{x}} P(\mathbf{x} \mid \mathcal{G}^*, \theta^*, z) \log P(\mathbf{x} \mid \mathcal{G}', \theta, z)$, which in general differs from the unconditional $\theta_{\mathcal{G}'}^\dagger$.

5.1. Assumptions. We organize the assumptions into four groups: causal assumptions (1–3), parametric assumptions (4–6), survey design assumptions (7–9), and a regularity assumption (10).

ASSUMPTION 1 (Causal Sufficiency). *All common causes of any pair (X_j, X_k) in $\mathbf{X} = (X_1, \dots, X_q)$ are included in the measured variables $\mathbf{X} \cup \mathbf{Z}$.*

ASSUMPTION 2 (Causal Markov Property). *The superpopulation distribution $P_\xi(\mathbf{X})$ factorizes with respect to the true causal DAG $\mathcal{G}^*$: $P_\xi(\mathbf{X}) = \prod_{j=1}^q P_\xi(X_j \mid \mathbf{X}_{\text{pa}(j)})$.*

ASSUMPTION 3 (Faithfulness). *Every conditional independence in P_ξ is entailed by the Markov property of $\mathcal{G}^*$: $X_A \perp\!\!\!\perp X_B \mid X_C$ in P_ξ if and only if A and B are d -separated by C in $\mathcal{G}^*$.*

ASSUMPTION 4 (Ordinal Regression Specification). *Each conditional distribution under the true DAG $\mathcal{G}^*$ follows a cumulative link model:*

$$P_\xi(X_j \leq \ell \mid \mathbf{X}_{\text{pa}(j)}, \mathbf{Z}) = F \left(\gamma_{j\ell} - \sum_{k \in \text{pa}(j)} \beta_{jk} X_k - \boldsymbol{\delta}_j' \mathbf{Z} - \alpha_j \right), \quad \ell = 1, \dots, L_j,$$

where F is a known, strictly increasing link function, $\gamma_{j1} < \dots < \gamma_{j,L_j} = \infty$ are threshold parameters with $\gamma_{j1} = 0$, β_{jk} are parent effect parameters with $\beta_{jk,1} = 0$, $\boldsymbol{\delta}_j$ are covariate adjustment parameters, and α_j is an intercept. We require $L_j \geq 3$ for all $j = 1, \dots, q$.

REMARK 4. *The restriction $L_j \geq 3$ is necessary: if $L_j = 2$, the cumulative link model collapses to a standard binary regression and causal directionality is identifiable only up to the Markov equivalence class (Ni and Mallick, 2022).*

ASSUMPTION 5 (Ordinal Identifiability). *Let $\mathcal{G}_1$ and $\mathcal{G}_2$ be any two Markov-equivalent DAGs on $\mathbf{X}$, and let θ_1 denote a parameter under $\mathcal{G}_1$ in the no-covariate oBN (Assumption 4 with $\boldsymbol{\delta}_j = \mathbf{0}$ for all j). Define the pseudo-true parameter $\theta_2^\dagger(\theta_1) = \arg \max_{\theta} \sum_{\mathbf{x}} P(\mathbf{x} \mid \mathcal{G}_1, \theta_1) \log P(\mathbf{x} \mid \mathcal{G}_2, \theta)$. Then the set of θ_1 for which $P(\mathbf{X} \mid \mathcal{G}_1, \theta_1) \equiv P(\mathbf{X} \mid \mathcal{G}_2, \theta_2^\dagger(\theta_1))$ is a proper analytic subvariety of $\Theta_{\mathcal{G}_1}$ of Lebesgue measure zero. We assume the true no-covariate parameter θ^* lies outside this exceptional set.*

Assumption 5 is the central identifiability result of Ni and Mallick (2022) for the bivariate oBN; its extension to multivariate DAGs follows from the pairwise decomposition argument of Ni, Chen and Wang (2025). Proposition 1 lifts this generic identifiability to the covariate-adjusted model.

ASSUMPTION 6 (Non-Degeneracy). *$P_\xi(X_j = \ell \mid \mathbf{X}_{\text{pa}(j)} = \mathbf{x}_{\text{pa}(j)}, \mathbf{Z} = \mathbf{z}) > 0$ for all j , all levels ℓ , all parent configurations $\mathbf{x}_{\text{pa}(j)}$, and all $\mathbf{z}$ in the support of $\mathbf{Z}$.*

ASSUMPTION 7 (Positivity). *Every unit $i \in U_N$ has a known, strictly positive first-order inclusion probability: $\pi_i = \Pr(i \in S_n) > 0$.*

ASSUMPTION 8 (Bounded Weights). *There exist constants $0 < c \leq C < \infty$, independent of N , such that $c \leq nw_i/N \leq C$ for all $i \in U_N$, ensuring that no single unit dominates the pseudo-likelihood.*

ASSUMPTION 9 (Design Consistency). *For any bounded, measurable function g on the joint sample space of $(\mathbf{X}, \mathbf{Z})$,*

$$\frac{1}{N} \sum_{i \in S_n} w_i g(\mathbf{x}_i, \mathbf{z}_i) - \frac{1}{N} \sum_{i \in U_N} g(\mathbf{x}_i, \mathbf{z}_i) \xrightarrow{p} 0$$

under the design distribution p_N , for each fixed realization of U_N .

Assumption 9 is satisfied by standard probability sampling designs under Assumptions 7–8; see Fuller (2009).

ASSUMPTION 10 (Regularity). *For each DAG $\mathcal{G}$: (i) the parameter space $\Theta_{\mathcal{G}}$ is compact; (ii) $\log P(\mathbf{x} \mid \mathcal{G}, \boldsymbol{\theta}, \mathbf{z})$ is twice continuously differentiable in $\boldsymbol{\theta}$ for each $(\mathbf{x}, \mathbf{z})$; (iii) the superpopulation expected log-likelihood $Q_{\xi}(\boldsymbol{\theta} \mid \mathcal{G}) = \mathbb{E}_{\xi}[\log P(\mathbf{X} \mid \mathcal{G}, \boldsymbol{\theta}, \mathbf{Z})]$ has a unique maximizer $\boldsymbol{\theta}_{\mathcal{G}}^*$ with nonsingular Hessian; and (iv) $\boldsymbol{\theta} \mapsto \log P(\mathbf{x} \mid \mathcal{G}, \boldsymbol{\theta}, \mathbf{z})$ satisfies a Lipschitz condition uniformly over $(\mathbf{x}, \mathbf{z})$.*

For the oBn with finitely many ordinal levels, condition (iv) is automatic since $\mathbf{X}$ has finite support and the link function F is smooth.

5.2. Main Results. Define the survey-weighted pseudo-log-likelihood under DAG $\mathcal{G}$ as

$$(9) \quad \tilde{\ell}(\boldsymbol{\theta} \mid \mathcal{G}) = \sum_{i \in S_n} w_i \log P(\mathbf{x}_i \mid \mathcal{G}, \boldsymbol{\theta}, \mathbf{z}_i),$$

the pseudo-MLE as $\hat{\boldsymbol{\theta}}_{\mathcal{G}}^w = \arg \max_{\boldsymbol{\theta}} \tilde{\ell}(\boldsymbol{\theta} \mid \mathcal{G})$, and the survey-weighted pseudo-BIC as

$$(10) \quad \widetilde{\text{pBIC}}(\mathcal{G}) = -2 \tilde{\ell}(\hat{\boldsymbol{\theta}}_{\mathcal{G}}^w \mid \mathcal{G}) + K_{\mathcal{G}} \cdot \rho_n,$$

where $K_{\mathcal{G}} = \sum_{j=1}^q (L_j - 1 + \sum_{k \in \text{pa}(j)} (L_k - 1) + \dim(\boldsymbol{\delta}_j))$ is the number of free parameters and ρ_n is a penalty sequence satisfying $\rho_n \rightarrow \infty$ and $\rho_n/N \rightarrow 0$. In practice, $\rho_n = \log(n_{\text{eff}})$.

PROPOSITION 1 (Identifiability under covariate adjustment). *Suppose Assumption 5 holds for the no-covariate oBn. Let $\mathcal{G}^*$ and $\mathcal{G}'$ be Markov-equivalent DAGs with $\mathcal{G}' \neq \mathcal{G}^*$. Then the set of $(\boldsymbol{\theta}^*, \boldsymbol{\delta}^*) \in \Theta_{\mathcal{G}^*} \times \mathbb{R}^{r_q}$ for which*

$$\mathbb{E}_{\mathbf{Z}} \left[\text{KL} \left(P_{\xi}(\mathbf{X} \mid \mathcal{G}^*, \boldsymbol{\theta}^*, \boldsymbol{\delta}^*, \mathbf{Z}) \parallel P(\mathbf{X} \mid \mathcal{G}', \boldsymbol{\theta}^{\dagger}(\mathbf{Z}), \mathbf{Z}) \right) \right] = 0$$

has Lebesgue measure zero. We assume throughout that the true parameters $(\boldsymbol{\theta}^, \boldsymbol{\delta}^*)$ of the data-generating model lie outside this exceptional set, in which case the expected KL divergence is strictly positive.*

PROPOSITION 2 (Pseudo-MLE Consistency). *Under Assumptions 4 and 6–10, for each DAG $\mathcal{G}$,*

$$\hat{\boldsymbol{\theta}}_{\mathcal{G}}^w \xrightarrow{p} \boldsymbol{\theta}_{\mathcal{G}}^*$$

under the joint $(\xi \times p_N)$ -probability, where $\boldsymbol{\theta}_{\mathcal{G}}^ = \arg \max_{\boldsymbol{\theta}} Q_{\xi}(\boldsymbol{\theta} \mid \mathcal{G})$.*

The proof of Proposition 2 proceeds in two stages, design consistency followed by super-population convergence (Fuller, 2009; Binder, 1983).

THEOREM 5.1 (Skeleton Consistency). *Under Assumptions 1–10, let $\mathcal{G}^*$ be the true DAG and $\mathcal{G}'$ any DAG not Markov-equivalent to $\mathcal{G}^*$. Then*

$$\Pr\left(\widehat{\text{pBIC}}(\mathcal{G}^*) < \widehat{\text{pBIC}}(\mathcal{G}')\right) \rightarrow 1.$$

Theorem 5.1 validates the addition and deletion moves under either scoring variant: the penalized pseudo-BIC correctly recovers the skeleton of $\mathcal{G}^*$. Within Markov equivalence classes, where the skeleton is fixed, orientation must be settled by a different mechanism: under heterogeneous L , single-BIC’s penalty terms do not cancel and introduce a directional bias toward low- L parents (Section 4.1). Split-BIC eliminates this bias by basing reversal moves on the unpenalized pseudo-likelihood. The next theorem shows this comparison is correct in the limit.

THEOREM 5.2 (Orientation Consistency). *Under Assumptions 1–10, let $\mathcal{G}^*$ be the true DAG and $\mathcal{G}'$ any DAG Markov-equivalent to $\mathcal{G}^*$ with $\mathcal{G}' \neq \mathcal{G}^*$. Then*

$$\Pr\left(\tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w | \mathcal{G}^*) > \tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w | \mathcal{G}')\right) \rightarrow 1.$$

Theorem 5.2 compares the unpenalized pseudo-likelihood under $\mathcal{G}^*$ to that under any Markov-equivalent $\mathcal{G}'$ and shows the correctly oriented DAG wins as $N \rightarrow \infty$. Split-BIC scores reversal moves by exactly this unpenalized pseudo-likelihood, so split-BIC’s reversal decisions converge to the correct orientation.

Single-BIC scores reversal moves by the penalized pseudo-BIC, which adds a term $(L_k - L_j)\rho_n$ to the comparison between $k \rightarrow j$ and $j \rightarrow k$. The level difference $|L_k - L_j|$ is fixed by the data and does not grow with N , so this added term is $O(\rho_n)$. Since $\rho_n/N \rightarrow 0$ by assumption, it is asymptotically dominated by the $O(N)$ likelihood gap, and single-BIC’s reversal decisions also converge to the correct orientation. Both variants therefore inherit orientation consistency.

COROLLARY 1 (Full DAG Recovery). *Under Assumptions 1–10, the swa-oBn algorithm under either scoring variant recovers the true DAG $\mathcal{G}^*$ with probability tending to one.*

6. Simulation Studies. We evaluate the finite-sample performance of swa-oBn through four simulation studies. Study 1 assesses the impact of survey design on DAG recovery; Study 2 examines covariate adjustment under confounding; Study 3 compares the single-BIC and split-BIC scoring variants of Section 4.1 across heterogeneous ordinal-level configurations. Study 4 demonstrates that the survey-weighting problem extends to other structure-learning algorithm families (PC, FCI, LiNGAM), with full results in the Supplementary Material.

6.1. Common Setup. All simulations share the following design. A finite population of $N = 20,000$ units is generated from an ordinal Bayesian network with $q = 8$ nodes, each having $L = 3$ ordinal levels except where noted. The true DAG is drawn once from an Erdős–Rényi model with edge probability 0.25 and held fixed across replications. Parent effects β_{jk} and intercepts α_j are drawn independently from $N(0, \sigma^2)$, where σ controls the signal strength of the cumulative link model (1).

From the population, a probability sample of approximate size n is drawn via Bernoulli sampling. Inclusion probabilities are assigned based on two designated oversampling variables (X_1 and X_q): each unit’s target score is $s_i = x_{i1} + x_{iq}$, and units in the top quartile of

s_i receive an elevated inclusion probability π_{hi} , with the rest receiving π_{lo} ($w_i = 1/\pi_i$). This two-stratum design induces informative sampling because the oversampling variables are nodes in the DAG. Three design tiers are considered, parameterized by the achieved design effect $\text{deff} = n/n_{\text{eff}}$: *none* (simple random sampling, $\text{deff} = 1$), *mild* ($\text{deff} \approx 1.5$), and *strong* ($\text{deff} \approx 3.2$). The strong tier is calibrated to approximate the design effect of the CAHPS analytic sample of Section 7 ($\text{deff} = 3.89$).

Each scenario is replicated 100 times. Performance is measured by the structural Hamming distance (SHD) between the estimated and true DAGs, defined as the number of edge additions, deletions, or reversals required to transform one into the other (de Jongh and Druzdzel, 2009). Edge precision and correct orientation rates are reported in the Supplementary Material. Single-BIC and split-BIC scoring of Section 4.1 coincide under the homogeneous $L = 3$ setting used in Studies 1 and 2; their comparison is the subject of Study 3.

6.2. Study 1: Effect of Survey Design on Structure Recovery. Study 1 compares swa-oBn with the unweighted oBn of Ni, Chen and Wang (2025) through three sub-studies, each varying one factor (Figure 1).

Panel (A) varies design strength at $n \approx 1,000$ and $\sigma = 1.5$. Under simple random sampling the methods coincide ($\text{SHD} = 2.94$ vs 2.93), confirming that subsequent differences are attributable to the weight correction. Under the mild design swa-oBn reduces SHD from 4.08 to 3.28; under the strong design the corresponding values are 4.23 and 3.80.

Panel (B) varies sample size under the strong design. Both methods improve with n , but the gap widens from 0.09 at $n = 300$ to 0.57 at $n = 4,000$. This pattern is the finite-sample signature of the bias characterized in Theorems 5.1 and 5.2: swa-oBn converges toward the truth while oBn plateaus above it.

Panel (C) varies signal strength. swa-oBn achieves lower SHD at every value of $\sigma \in \{0.25, 0.75, 1.0, 1.5, 2.0\}$ and is statistically indistinguishable at $\sigma = 0.5$. The largest gap occurs at $\sigma = 1.5$ ($\text{SHD} = 4.16$ vs 3.70), where the ordinal identifiability mechanism retains sufficient signal to orient edges while design-induced bias remains strong enough to mislead the unweighted method.

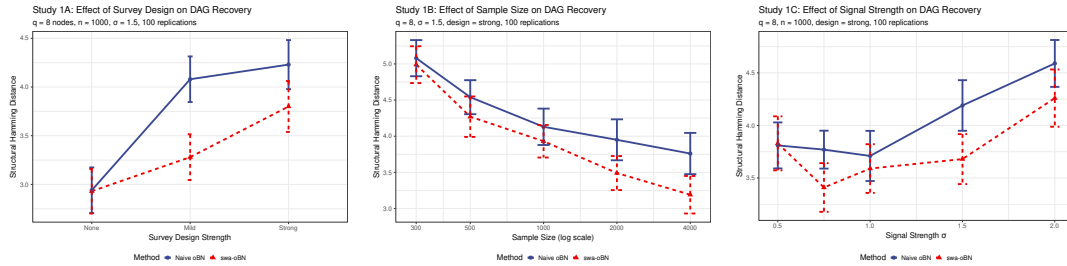


FIG 1. Mean SHD (± 1.96 SE) for swa-oBn and oBn of Ni, Chen and Wang (2025) across 100 replications. (A) Design strength at $n \approx 1,000$, $\sigma = 1.5$. (B) Sample size under the strong design, $\sigma = 1.5$. (C) Signal strength under the strong design at $n \approx 1,000$. All panels: $q = 8$, $L = 3$.

6.3. Study 2: Value of Covariate Adjustment. Study 2 isolates the role of covariate adjustment and its interaction with survey weighting. The true DAG has $q = 8$ nodes partitioned into six upstream nodes and two terminal outcomes, with two parallel chains $X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow X_4$ and $X_1 \rightarrow X_5 \rightarrow X_6$ emanating from a common root; outcome X_7 has parent X_3 , and outcome X_8 is a collider with parents X_4 and X_6 . A confounder $Z \sim N(0, 1)$ enters the cumulative link model (7) additively for nodes $\{X_2, X_3, X_5, X_7\}$ with common effect

magnitude δ . The confounded set is disjoint from the oversampling variables $\{X_1, X_8\}$ so that survey-design and confounding biases are mechanically independent. Four estimators are compared: oBn (no weights, no $\mathbf{Z}$), $\mathbf{Z}$ -adjusted (no weights, $\mathbf{Z}$ included), *weighted* (weights, no $\mathbf{Z}$), and swa-oBn (weights and $\mathbf{Z}$).

Figure 2(A) varies δ at the strong design with $n \approx 1,000$ and $\sigma = 1.5$. At $\delta = 0$ the four estimators are statistically indistinguishable. As δ grows, oBn and the weighted estimator degrade in parallel, reaching $\text{SHD} = 7.59$ and 7.33 at $\delta = 2.0$, confirming that weighting alone does not address confounding. The $\mathbf{Z}$ -adjusted and swa-oBn estimators improve with δ , falling to 5.03 and 5.24 respectively.

Figure 2(B) fixes $\delta = 1.2$ and varies the design. Under simple random sampling, $\mathbf{Z}$ -adjustment and swa-oBn coincide ($\text{SHD} = 4.10$ each): weights are vacuous when sampling is uninformative. Under the mild design swa-oBn attains the lowest SHD (4.57 vs 5.31 for $\mathbf{Z}$ -adjusted, 6.69 for weighted, 7.17 for naive). Under the strong design swa-oBn remains best (5.77 vs 5.92 for $\mathbf{Z}$ -adjusted, 7.08 for weighted, 7.79 for naive), though the margin over $\mathbf{Z}$ -adjustment narrows. Survey weighting and covariate adjustment thus address distinct bias sources, with their joint application offering the largest gains under intermediate design strength.

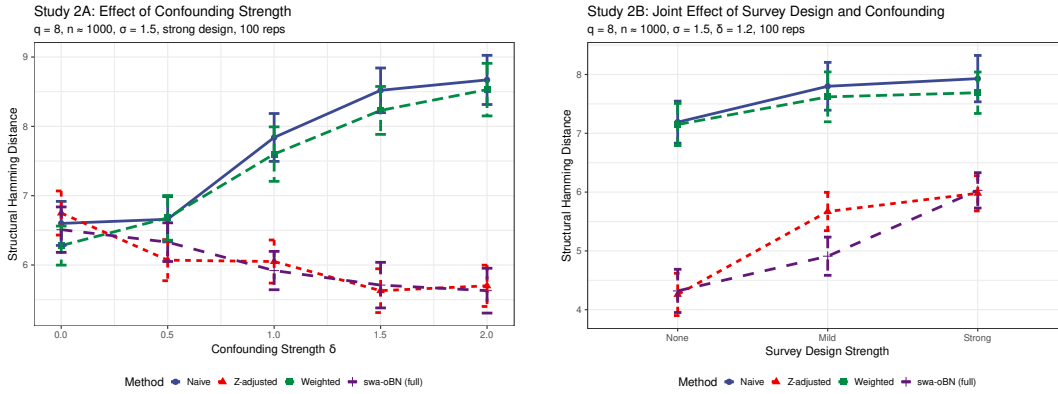


FIG 2. Mean SHD (± 1.96 SE) for four estimators across 100 replications. (A) Confounding strength δ under the strong design ($\text{deff} \approx 3.3$); weighting alone fails to address confounding, while $\mathbf{Z}$ -adjustment and swa-oBn improve with δ . (B) Design strength with $\delta = 1.2$ fixed; swa-oBn is best at every design tier and ties with $\mathbf{Z}$ -adjustment when sampling is uninformative. Both panels: $q = 8, \sigma = 1.5$.

6.4. Study 3: Single-BIC versus Split-BIC Scoring. Study 3 evaluates the two scoring variants of Section 4.1 across heterogeneous ordinal-level configurations. The true DAG has $q = 10$ nodes; the level vector $\mathbf{L} = (L_1, \dots, L_{10})$ varies across six regimes (Table 1). Each regime is replicated 100 times at $n \in \{200, 400, 700, 1,000\}$ under the strong design, with four paired fits per replication: split-BIC and single-BIC, each with and without survey weighting. Table 1 reports the paired SHD gap at $n = 1,000$ under survey weighting. Split-BIC outperforms single-BIC in five regimes, with relative gains up to 19.7% under *Wide*. The exception is *Graded* (+12.7% in favor of single-BIC). The mechanism is the per-edge orientation bias of the unpenalized second pass: under sharply separated or homogeneous $\mathbf{L}$, biases accumulate coherently and split-BIC dominates; when intermediate values populate the range, biases offset and single-BIC's regularization wins. Supplement 5 confirms the taxonomy is stable across n . The CAHPS analytic sample has $\mathbf{L} = (3, 3, 3, 3, 3, 3, 3, 3, 4, 5)$, placing it in the *Graded* regime; Section 7 therefore uses single-BIC. The regime taxonomy

applies independently of survey weighting and is therefore a contribution to ordinal Bayesian network learning more broadly (Supplement 5).

TABLE 1
Paired SHD gap (split-BIC – single-BIC) under survey weighting at $n = 1,000$. Negative values favor split-BIC. All gaps are significant at the $|\text{gap}| > 1.96 \cdot \text{SE}$ threshold.

Regime	L -set	Single-BIC	Split-BIC	Gap (SE)	Relative
Uniform	$\{3\}$	4.12	4.05	−0.07 (0.02)	−1.7%
Adjacent	$\{3, 4\}$	4.54	4.08	−0.46 (0.06)	−10.2%
Graded	$\{3, 4, 5\}$	3.32	3.74	+0.42 (0.04)	+12.7%
Polarized	$\{3, 5\}$	4.33	3.72	−0.61 (0.05)	−14.1%
Wide	$\{3, 7\}$	4.97	3.99	−0.98 (0.07)	−19.7%
Extreme	$\{3, 9\}$	4.87	4.06	−0.81 (0.06)	−16.6%

6.5. *Study 4: Generality of the Survey-Weighting Problem.* To confirm that the distortion introduced by informative sampling is not specific to ordinal Bayesian networks, we applied survey-weighted and unweighted versions of three additional algorithms: the PC algorithm, FCI, and LiNGAM, to a simple $X \rightarrow Y \rightarrow Z$ chain. Under mild and strong designs, the unweighted versions frequently infer a spurious $X-Z$ edge, while their weighted counterparts recover the correct structure. Complete results appear in the Supplementary Material.

7. Application: Causal Architecture of Veteran Social Determinants of Health. We apply the `swa-oBn` to the VHA CAHPS data of Section 2 to address the scientific questions posed therein. Sensitivity analyses concerning the proportional odds working assumption are deferred to the Supplementary Material.

7.1. *Point-estimate DAG and bootstrap uncertainty.* We ran `swa-oBn` with $R = 10$ random restarts, the covariates Z of Section 2.4, and the outcome-to-need blacklist. Kish’s effective sample size is $n_{\text{eff}} = 1,387$ (design effect 3.89), so the pseudo-BIC penalty scales with $\log n_{\text{eff}} \approx 7.23$. Uncertainty was quantified by $B = 250$ nonparametric pairs bootstrap resamples (Section 4.4).

Figure 3 displays the bootstrap median-probability-model (MPM) DAG (Barbieri and Berger, 2004), retaining directed edges with inclusion probability $\hat{p}_{jk} \geq 0.50$ and classifying them as *robust* ($\hat{p}_{jk} \geq 0.75$) or *suggestive* ($0.50 \leq \hat{p}_{jk} < 0.75$). Table 2 reports the 14 consensus edges (8 robust, 6 suggestive); the full 10×10 matrix appears in Supplementary Table 5. Nine of ten nodes are connected; *childcare* is isolated, reflecting low prevalence of childcare needs in an older veteran sample. The edge between mental health and well-being is present in every resample but its orientation is not identified ($\hat{p}_{\text{WB} \rightarrow \text{MH}} = 0.54$ vs. $\hat{p}_{\text{MH} \rightarrow \text{WB}} = 0.46$), as expected for cross-sectional data on simultaneously-determined outcomes.

7.2. *Weighted versus unweighted comparison.* To assess the impact of survey weighting, we re-ran the analysis with all weights set to one, holding all other settings fixed. The unweighted analysis retains 23 consensus edges versus 14 weighted, of which 11 are shared (3 with reversed direction), 12 are only-unweighted, and 3 are only-weighted. The side-by-side visualization is given in Supplementary §11; the full edge-level breakdown is the 10×10 matrix in Supplementary Table 5.

The substantive difference concerns discrimination/isolation. Its out-degree is 4 under weighting, placing it as a proximal mediator downstream of basic needs; under no weighting

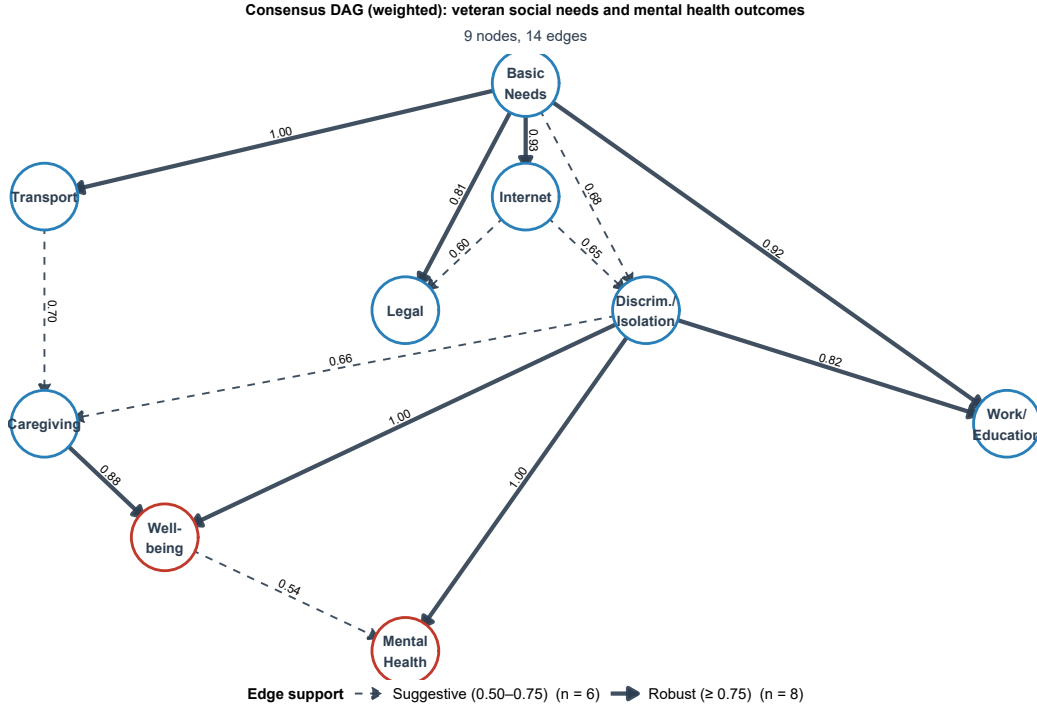


FIG 3. Consensus DAG for veteran social needs and mental health outcomes, estimated by *swa-on* on the VA SHEP January–February 2023 primary care sample ($n = 5,396$), adjusted for age and race-ethnicity-sex and weighted by SHEP sampling weights. The self-reported social need nodes in the figure include: (1) basic needs (i.e., paying for basics like food, housing, medical care, and heating; paying for food (only); getting or maintaining housing); (2) work/education (finding or keeping work, getting additional education or job training); (3) discrimination/isolation (managing experiences of discrimination; feeling social isolated; feeling lonely); (4) caregiving (adult caregiving for self or others); (5) transportation (for basic needs like medical care or grocery shopping); (6) internet (getting internet at home); (7) legal (obtaining assistance for legal issues). Childcare was not connected to other nodes; thus, it is not shown in the figure. Outcome-to-need edges forbidden by blacklist. Edge probabilities are bootstrap inclusion frequencies over $B = 250$ resamples (all converged). Displayed edges have $\hat{p}_{jk} \geq 0.50$; dashed edges are suggestive ($\hat{p} \in [0.50, 0.75]$), solid are robust ($\hat{p} \geq 0.75$).

it rises to 8, placing it as the root of the entire network. Non-White-male veterans were over-sampled by the SHEP design and have elevated discrimination-related needs (Frank et al., 2025; Russell et al., 2026), so unweighted estimation reads the resulting sample-level correlations as a common cause; weighting back to the superpopulation reveals that discrimination/isolation is structurally a mediator. An unweighted analysis would conclude that material deprivation acts directly on well-being rather than through discrimination/isolation, a qualitatively different policy conclusion. The pattern is consistent with the simulation evidence of Section 6.2.

7.3. Substantive findings. The weighted consensus DAG yields three findings addressing the scientific questions of Section 2.

Unmet basic needs are the upstream hub. Basic needs has direct edges to five of the seven other need nodes to which it can connect, with inclusion probabilities from 0.68 to 1.00; no other need variable has comparable out-degree. Basic financial deprivation is therefore the root cause in the social-need network, and interventions addressing food, housing, and basic necessities have the greatest potential for cascading downstream benefits. This interpretation

TABLE 2

Bootstrap inclusion probabilities for the 14 consensus edges of Figure 3 ($B = 250$). The self-reported social need nodes in the table include: (1) basic needs (i.e., paying for basics like food, housing, medical care, and heating; paying for food (only); getting or maintaining housing); (2) work/education (finding or keeping work, getting additional education or job training); (3) discrimination/isolation (managing experiences of discrimination; feeling social isolated; feeling lonely); (4) caregiving (adult caregiving for self or others); (5) transportation (for basic needs like medical care or grocery shopping); (6) internet (getting internet at home); (7) legal (obtaining assistance for legal issues). Childcare was not connected to other nodes; thus, it is not shown in the table. Robust: $\hat{p}_{jk} \geq 0.75$; suggestive: $\hat{p}_{jk} \in [0.50, 0.75)$. Full 10×10 matrix in Supplementary Table 5.

From	To	$\hat{p}_{jk}$	Tier
Discrimination/Isolation	Mental Health	1.00	Robust
Basic Needs	Transport	1.00	Robust
Discrimination/Isolation	Well-being	1.00	Robust
Basic Needs	Internet	0.93	Robust
Basic Needs	Work/Education	0.92	Robust
Caregiving	Well-being	0.88	Robust
Discrimination/Isolation	Work/Education	0.82	Robust
Basic Needs	Legal	0.81	Robust
Transport	Caregiving	0.70	Suggestive
Basic Needs	Discrim./Isolation	0.68	Suggestive
Discrimination/Isolation	Caregiving	0.66	Suggestive
Internet	Discrim./Isolation	0.65	Suggestive
Internet	Legal	0.60	Suggestive
Well-being	Mental Health	0.54	Suggestive

is conditional on causal sufficiency; unmeasured confounders such as income or geography could alter the estimated structure.

Discrimination and isolation are the proximal mediators of health. Edges from discrimination/isolation to well-being and mental health both carry $\hat{p} = 1.00$, the highest in the graph; most individual needs have no direct edge to either outcome. Material deprivation thus reaches health primarily through discrimination and isolation, so interventions targeting material needs alone may fail to improve outcomes. As shown in Section 7.2, this finding is conditional on appropriate survey weighting.

Mental health is reached through mediation, not directly. Mental health receives no direct edges from individual needs; the primary pathway is needs $\rightarrow$ discrimination/isolation $\rightarrow$ mental health, with a secondary contribution via caregiving $\rightarrow$ well-being $\rightarrow$ mental health. The caregiving $\rightarrow$ well-being edge ($\hat{p} = 0.88$) is noteworthy because caregiving burden is not among the nine social risk domains assessed by the VHA’s ACORN screening initiative (Russell et al., 2023) or the five core domains of the CMS Accountable Health Communities HRSN tool (Billieux et al., 2017), yet the DAG identifies it as a proximal determinant of well-being.

8. Discussion. This paper introduced the survey-weighted, covariate-adjusted ordinal Bayesian network (swa-oBn), a framework for causal discovery from complex survey data with ordinal responses. The method integrates Horvitz–Thompson weights and Kish’s effective sample size into a pseudo-BIC scoring framework, presents two co-equal scoring variants (single-BIC and split-BIC) with a regime-based selection rule for heterogeneous ordinal scales, supports exogenous covariate adjustment without expanding the DAG, and allows domain knowledge to be encoded via structural constraints. We proved that the framework con-

sistently recovers the true population-level causal graph under a joint superpopulation-and-design asymptotic framework: skeleton consistency follows from the penalized pseudo-BIC under either variant, and orientation consistency is inherited from generic ordinal identifiability.

Applied to the VHA CAHPS survey, the `swa-OBn` revealed a causal architecture of Veteran social determinants of health that could not have been discovered by the associational methods used in prior analyses of this dataset (Frank et al., 2025; Lamba et al., 2025; Russell et al., 2026). Three principal findings emerged. First, unmet basic needs (food, housing, paying for basics) function as the upstream causal hub, with direct edges to five of the seven other need nodes. Second, discrimination and social isolation serve as the proximal mediator of health outcomes, with bootstrap inclusion probabilities of 1.00 for both the well-being and mental health edges. Third, mental health is reached through mediation, primarily via discrimination/isolation and degraded well-being, rather than directly from most individual needs. These findings provide the directional knowledge required for intervention prioritization: addressing basic needs has the greatest potential for cascading downstream benefits, but interventions that do not also address discrimination and isolation may fail to reach health outcomes through the primary mediating channel.

The demographic covariates Z (age and the combined race-ethnicity-sex indicator) are the same variables adjusted for in all three companion analyses of this dataset, where they substantially altered prevalence ratios (Frank et al., 2025) and risk-need concordance patterns (Russell et al., 2026). Age captures cohort and life-stage effects; the race-ethnicity-sex composite absorbs the systematic disparities documented in those studies and reflects the stratified oversampling design. These are plausibly the dominant demographic confounders for the relationships among social need variables in this population. However, several variables that could confound need-need or need-outcome relationships remain unmeasured in the CAHPS data. We do not claim that causal sufficiency holds perfectly. Rather, we argue that the Z -adjusted DAG is more credible than the unadjusted one and that the methodology is correct conditional on the assumption. If richer covariate data become available in future CAHPS administrations, they can be incorporated into the `swa-OBn` framework with no algorithmic changes; only the Z matrix needs to be expanded.

Several additional limitations merit discussion. First, the greedy hill-climbing search is guaranteed to find only a local optimum. The multi-start procedure with $R = 10$ random initializations mitigates this concern, and the bootstrap procedure implicitly explores the DAG space by re-running the search on resampled data; nevertheless, the estimated DAG is not guaranteed to be the global optimum. Second, the polychoric clustering used to construct composite nodes introduces a working-model approximation. The pointwise maximum across constituent three-level items, while still a three-level ordinal, does not in general satisfy an exact cumulative link specification conditional on its parents, so the identifiability guarantee of Ni and Mallick (2022) holds only approximately for these composite nodes. We treat this as a deliberate trade-off: the alternative (entering all 15 raw items into the graph) produces an overly dense and substantively uninterpretable structure. The bootstrap procedure provides some robustness against orientation artifacts arising from this approximation. Third, the global effective sample size n_{eff} used in the pseudo-BIC penalty does not capture node-specific design effects. Different variables may have different design effects depending on how strongly the oversampling correlates with each variable. Incorporating node-specific design effects would require estimating variable-level design effects and integrating them into the scoring, an extension we leave for future work. Fourth, the naive nonparametric bootstrap does not exactly replicate the variance structure of the stratified sampling design; a design-consistent alternative such as the rescaled bootstrap of Rao and Wu (1988) would improve the calibration of intermediate-valued edge inclusion probabilities. Fifth, the CAHPS data

are cross-sectional, so the discovered DAG represents a contemporaneous causal structure, not a temporal one. The blacklist encoding the assumption that outcomes do not cause needs is the minimal temporal constraint imposed; all other directionality is learned from the data. Sixth, the proportional odds structure of the cumulative link model is treated as an explicitly designated working assumption rather than a tested property of the data; the Supplementary Material documents diagnostic checks and the anti-conservatism of standard diagnostics under complex survey designs.

The survey-weighting problem documented in this paper is not specific to ordinal Bayesian networks. The Supplementary Material demonstrates that the PC algorithm, FCI, and LiNGAM all suffer from analogous distortions when applied to complex survey data. Developing fully survey-weighted extensions of these algorithms for multivariate settings, with formal consistency guarantees comparable to those established here for the `swa-oBn`, is a natural next step and would provide practitioners with a complete toolkit for design-consistent causal discovery. A further methodological priority is sensitivity analysis for unmeasured confounding in causal discovery; adapting E-value methods (VanderWeele and Ding, 2017) from treatment-effect estimation to a graph-valued target would provide a principled framework for assessing robustness of the learned DAG. The split-BIC variant, which decouples skeleton control from orientation control, may also be of independent methodological interest for ordinal Bayesian network learning beyond the survey-weighted setting; a fuller theoretical and empirical treatment of the regime-based choice between split-BIC and single-BIC could be developed as a standalone contribution. Finally, longitudinal extensions of the CAHPS survey, if available, would enable temporal causal discovery and permit relaxation of the cross-sectional assumption that currently limits the interpretation of the discovered graph.

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SUPPLEMENTARY MATERIAL

Supplementary Material for “Causal Discovery from Complex Survey Data”

Full proofs of Propositions 1–2 and Theorems 5.1–5.2; the `swa-oBn` algorithm in pseudocode; polychoric clustering details; the composite node table; Study 1 supplementary metrics; Study 3 supplementary tables; Study 4 results (survey-weighted PC, FCI, LiNGAM); complete bootstrap edge probability matrix; weighted-versus-unweighted comparison; proportional odds diagnostics; survey weight distribution details.

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